

Homework 2

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①

$$\begin{aligned} h_{\text{MAP}} &= \underset{h}{\operatorname{argmax}} P(\kappa | h) P(h) = \\ &= \underset{h}{\operatorname{argmax}} P((y_1, y_2) | y_6 = h) P((y_3, y_4) | y_6 = h) P(y_5 | y_6 = h) P(y_6 = h) \end{aligned}$$

$$\boxed{(y_1, y_2)}$$

$$y_6 = N$$

$$X = \{(0,52; 0,80), (0,53; 0,92), (0,42; 0,48)\}$$

$$\mu_N = \frac{1}{3} \left(\begin{bmatrix} 0,52 \\ 0,80 \end{bmatrix} + \begin{bmatrix} 0,53 \\ 0,92 \end{bmatrix} + \begin{bmatrix} 0,42 \\ 0,48 \end{bmatrix} \right) = \begin{bmatrix} 0,49 \\ 0,7333 \end{bmatrix}$$

$$\begin{aligned} \sum_{N y_1 y_1} &= \frac{1}{3-1} \left[(0,52-0,49)^2 + (0,53-0,49)^2 + (0,42-0,49)^2 \right] = \\ &= 0,0037 \end{aligned}$$

$$\begin{aligned} \sum_{N y_1 y_2} &= \frac{1}{3-1} \left[(0,52-0,49)(0,80-0,7333) + (0,53-0,49)(\overset{0,92}{\cancel{0,80}}-0,7333) + \right. \\ &\quad \left. + (0,42-0,49)(0,48-0,7333) \right] = \\ &= 0,0136 \end{aligned}$$

$$\sum_{N y_2 y_2} = \frac{1}{3-1} \left[(0,80-0,7333)^2 + (0,92-0,7333)^2 + (0,48-0,7333)^2 \right] =$$

$$\Sigma_N = \begin{bmatrix} 0,0037 & 0,0136 \\ 0,0136 & 0,0517 \end{bmatrix} \quad |\Sigma_N| = 6,33 \times 10^{-6}$$

$$y_6 = P$$

$$X = \{(0,49; 0,58), (0,62; 0,31), (0,44; 0,38)\}$$

$$\mu_p = \frac{1}{3} \left(\begin{bmatrix} 0,49 \\ 0,58 \end{bmatrix} + \begin{bmatrix} 0,62 \\ 0,31 \end{bmatrix} + \begin{bmatrix} 0,44 \\ 0,38 \end{bmatrix} \right) = \begin{bmatrix} 0,5167 \\ 0,4233 \end{bmatrix}$$

$$\Sigma_{p y_1 y_1} = \frac{1}{3-1} \left[(0,49 - 0,5167)^2 + (0,62 - 0,5167)^2 + (0,44 - 0,5167)^2 \right] = 0,0086$$

$$\Sigma_{p y_1 y_2} = \frac{1}{3-1} \left[(0,49 - 0,5167)(0,58 - 0,4233) + (0,62 - 0,5167)(0,31 - 0,4233) + (0,44 - 0,5167)(0,38 - 0,4233) \right] = -0,0063$$

$$\Sigma_{p y_2 y_2} = \frac{1}{3-1} \left[(0,58 - 0,4233)^2 + (0,31 - 0,4233)^2 + (0,38 - 0,4233)^2 \right] = 0,0196$$

$$\Sigma_P = \begin{bmatrix} 0,0086 & -0,0063 \\ -0,0063 & 0,0196 \end{bmatrix} \quad |\Sigma_P| = 1,2887 \times 10^{-4}$$

$$(y_3, y_4)$$

$$y_6 = N$$

$$x = \{(0,1), (0,0), (0,1)\}$$

$$P((y_3, y_4) = (0,0) | y_6 = N) = 1/3$$

$$P((y_3, y_4) = (0,1) | y_6 = N) = 2/3$$

$$P((y_3, y_4) = (1,0) | y_6 = N) = 0$$

$$P((y_3, y_4) = (1,1) | y_6 = N) = 0$$

$$y_6 = P$$

$$x = \{(1,0), (0,0), (0,0)\}$$

$$P((y_3, y_4) = (0,0) | y_6 = P) = 2/3$$

$$P((y_3, y_4) = (0,1) | y_6 = P) = 0$$

$$P((y_3, y_4) = (1,0) | y_6 = P) = 1/3$$

$$P((y_3, y_4) = (1,1) | y_6 = P) = 0$$

$$y_5$$

$$y_6 = N$$

$$x = \{1, 0, 1\}$$

$$P(y_5 = 0 | y_6 = N) = 1/3$$

$$P(y_5 = 1 | y_6 = N) = 2/3$$

$$y_6 = P$$

$$x = \{1, 1, 1\}$$

$$P(y_5 = 0 | y_6 = P) = 0$$

$$P(y_5 = 1 | y_6 = P) = 1$$

$$\text{Priors}$$

$$P(y_6 = N) = \frac{3}{6} = \frac{1}{2}$$

$$P(y_6 = P) = \frac{1}{2}$$

b)

$$\mu_7 = [0,45 \quad 0,80 \quad 0 \quad 0 \quad 1]^T$$

$$P((y_1, y_2) = (0,45; 0,80) | y_6 = N) =$$

$$= \frac{1}{2\pi \sqrt{6,33 \times 10^{-6}}} \exp\left(-\frac{1}{2} \left(\begin{bmatrix} 0,45 \\ 0,80 \end{bmatrix} - \begin{bmatrix} 0,49 \\ 0,7333 \end{bmatrix} \right)^T \cdot \Sigma_N^{-1} \cdot \left(\begin{bmatrix} 0,45 \\ 0,80 \end{bmatrix} - \begin{bmatrix} 0,49 \\ 0,7333 \end{bmatrix} \right)\right) =$$

$$= 0,000104$$

$$P((y_1, y_2) = (0,45; 0,80) | y_6 = P) =$$

$$= \frac{1}{2\pi \sqrt{|\Sigma_P|}} \exp\left(-\frac{1}{2} \left(\begin{bmatrix} 0,45 \\ 0,80 \end{bmatrix} - \mu_P \right)^T \cdot \Sigma_P^{-1} \cdot \left(\begin{bmatrix} 0,45 \\ 0,80 \end{bmatrix} - \mu_P \right)\right) =$$

$$= 0,302295$$

$$P((y_3, y_4) = (0,0) | y_6 = N) = 1/3$$

$$P((y_3, y_4) = (0,0) | y_6 = P) = 2/3$$

$$P(y_5 = 1 | y_6 = N) = 2/3$$

$$P(y_5 = 1 | y_6 = P) = 1$$

$$P((y_1, y_2) = (0,45; 0,80) | y_6 = N) P((y_3, y_4) = (0,0) | y_6 = N) P(y_5 = 1 | y_6 = N) P(y_6 = N) =$$

$$= 1,15 \times 10^{-5}$$

$$P((y_1, y_2) = (0,45; 0,80) | y_6 = P) P((y_3, y_4) = (0,0) | y_6 = P) P(y_5 = 1 | y_6 = P) P(y_6 = P) =$$

$$= 0,100765$$

$$\Rightarrow \ln_{MAP}(\mu_7) = P$$

$$\kappa_8 = [0,50 \quad 0,70 \quad 0 \quad 1 \quad 1]^T$$

$$P((y_1, y_2) = (0,50; 0,70) | y_6 = N) =$$

$$= \frac{1}{2\pi \sqrt{|\Sigma_N|}} \exp\left(-\frac{1}{2} \left(\begin{bmatrix} 0,50 \\ 0,70 \end{bmatrix} - \mu_N\right)^T \cdot \Sigma_N^{-1} \cdot \left(\begin{bmatrix} 0,50 \\ 0,70 \end{bmatrix} - \mu_N\right)\right) =$$

$$= 15,115668$$

$$P((y_1, y_2) = (0,50; 0,70) | y_6 = P) =$$

$$= \frac{1}{2\pi \sqrt{|\Sigma_P|}} \exp\left(-\frac{1}{2} \left(\begin{bmatrix} 0,50 \\ 0,70 \end{bmatrix} - \mu_P\right)^T \cdot \Sigma_P^{-1} \cdot \left(\begin{bmatrix} 0,50 \\ 0,70 \end{bmatrix} - \mu_P\right)\right) =$$

$$= 1,345298$$

$$P((y_1, y_2) = (0,50; 0,70) | y_6 = N) P((y_3, y_4) = (0,1) | y_6 = N) P(y_5 = 1 | y_6 = N) P(y_6 = N) =$$

$$= 15,115668 \times \frac{2}{3} \times \frac{2}{3} \times \frac{1}{2} = 3,35904$$

$$P((y_1, y_2) = (0,50; 0,70) | y_6 = P) P((y_3, y_4) = (0,1) | y_6 = P) P(y_5 = 1 | y_6 = P) P(y_6 = P) =$$

$$= 1,345298 \times 0 \times 1 \times \frac{2}{2} = 0$$

$$\Rightarrow h_{\text{MAP}}(\kappa_8) = N$$

(2)

$$x_1 \rightarrow 0 \ 1 \ 1$$

D	y_3	y_4	y_5	Hamming distance
x_2	0	0	0	2
x_3	0	1	1	0 ←
x_4	1	0	1	2
x_5	0	0	1	1
x_6	0	0	1	1
x_7	0	0	1	1
x_8	0	1	1	0 ←

There is no radius containing only 3 neighbours, so we'll consider $k=2$.

$x_3: N$
 $x_8: N$

} → $x_1: N$

x_1 , x_3 and x_8 have the same features, so Hamming distance results will be the same.

Thus x_1 , x_3 and x_8 will all be classified as N.

$$x_2 \rightarrow 0 \ 0 \ 0$$

D	y_3	y_4	y_5	Hamming distance
x_1	0	1	1	2
x_3	0	1	1	2
x_4	1	0	1	2
x_5	0	0	1	1 ←
x_6	0	0	1	1 ←
x_7	0	0	1	1 ←
x_8	0	1	1	2

$x_5: P$
 $x_6: P$
 $x_7: P$

} → $x_2: P$

$$|N_4 \rightarrow 1 \ 0 \ 1|$$

D	y ₃	y ₄	y ₅	Hamming distance
N ₁	0	1	1	2
N ₂	0	0	0	2
N ₃	0	1	1	2
N ₅	0	0	1	1 ←
N ₆	0	0	1	1 ←
N ₇	0	0	1	1 ←
N ₈	0	1	1	2

N₅: P
 N₆: P
 N₇: P

} → N₄: P

$$|N_5 \rightarrow 0 \ 0 \ 1|$$

D	y ₃	y ₄	y ₅	Hamming distance
N ₁	0	1	1	1
N ₂	0	0	0	1
N ₃	0	1	1	1
N ₄	0	0	1	1
N ₆	0	0	1	0 ←
N ₇	0	0	1	0 ←
N ₈	0	1	1	1

There is no radius containing only 3 neighbours, so we'll consider $k=2$

N₆: P
 N₇: P

} → N₅: P

N₅, N₆ and N₇ have the same features, so Hamming distance results will be the same.

Thus, N₅, N₆ and N₇ will all be classified as P.

D	Real	Predicted
N ₁	N	N
N ₂	N	P
N ₃	N	N
N ₄	P	P
N ₅	P	P
N ₆	P	P
N ₇	P	P
N ₈	N	N

$$\text{accuracy} = \frac{\text{hits}}{\text{attempts}} = \frac{7}{8} = 87,5\%$$

③

a)

$$\theta_{\text{Bayes}} = \underset{\theta}{\operatorname{argmax}} P(x|\theta) P(\theta)$$

$$P(x=x|\theta=0) = P(x=x|\theta=1)$$

$\Downarrow$

$$\theta_{\text{Bayes}} = \underset{\theta}{\operatorname{argmax}} P(\theta)$$

It is said that $P(\theta=0)=p$ and $p \in [1/2, 1]$,
so $\theta_{\text{Bayes}} = 0$

$$E_{\text{Bayes}} = P(\theta \neq \theta_{\text{Bayes}}) = P(\theta \neq 0) = 1-p$$

b)

$$\begin{aligned} P(x=x) &= P(x=x|\theta=0) P(\theta=0) + P(x=x|\theta=1) P(\theta=1) = \\ &= P(x=x|\theta=0) \cdot (P(\theta=0) + P(\theta=1)) = \\ &= P(x=x|\theta=0) \cdot (p + 1-p) = \\ &= P(x=x|\theta=0) \end{aligned}$$

$$\begin{aligned} 2 \times P(\theta=0|x=x) \times P(\theta=1|x=x) &= \\ = 2 \times \frac{P(x=x|\theta=0) P(\theta=0)}{P(x=x)} \times \frac{P(x=x|\theta=1) P(\theta=1)}{P(x=x)} &= \\ = 2 \times \frac{P(x=x)^2 P(\theta=0) \times P(\theta=1)}{P(x=x)^2} &= 2p(1-p) \end{aligned}$$

c)

$$E_{NN} = 2p(1-p) =$$

$$\stackrel{E_{\text{Bayes}} = 1-p}{=} 2 \times (1 - E_{\text{Bayes}}) \times E_{\text{Bayes}} =$$

$$= 2 \times E_{\text{Bayes}} \times (1 - E_{\text{Bayes}})$$